



Shri Vile Parle Kelavani Mandal's

DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

(Autonomous College Affiliated to the University of Mumbai)

NAAC Accredited with "A" Grade (CGPA : 3.18)



Name: Gargi Sukhatankar

Branch: CSE (ICB)

Subject: Cloud Computing Laboratory

SAP ID: 60019220125

Batch: B1

EXPERIMENT NO. 15

Aim: To study and implement container orchestration using Kubernetes on Google cloud platform

Theory Introduction to Container Orchestration and Kubernetes

Container orchestration refers to the automated arrangement, coordination, and management of software containers. As organizations increasingly adopt microservices architecture and containerization (e.g., using Docker), managing containers at scale becomes challenging. Kubernetes, originally developed by Google and now maintained by the Cloud Native Computing Foundation (CNCF), is the most widely adopted container orchestration platform.

What is Kubernetes?

Kubernetes (often abbreviated as K8s) is an open-source platform designed to automate the deployment, scaling, and operation of application containers. It provides a framework to run distributed systems resiliently. Kubernetes handles service discovery, load balancing, storage orchestration, self-healing, and automated rollouts and rollbacks.

Key Components of Kubernetes

- **Pods:** The smallest deployable units in Kubernetes that can contain one or more containers.
- **Nodes:** The physical or virtual machines on which Kubernetes runs workloads.
- **Clusters:** A set of nodes managed by the Kubernetes control plane.
- **Deployments:** Objects that define the desired state for Pods and ReplicaSets.
- **Services:** Abstract way to expose an application running on a set of Pods.

Why Use Kubernetes on Google Cloud Platform?

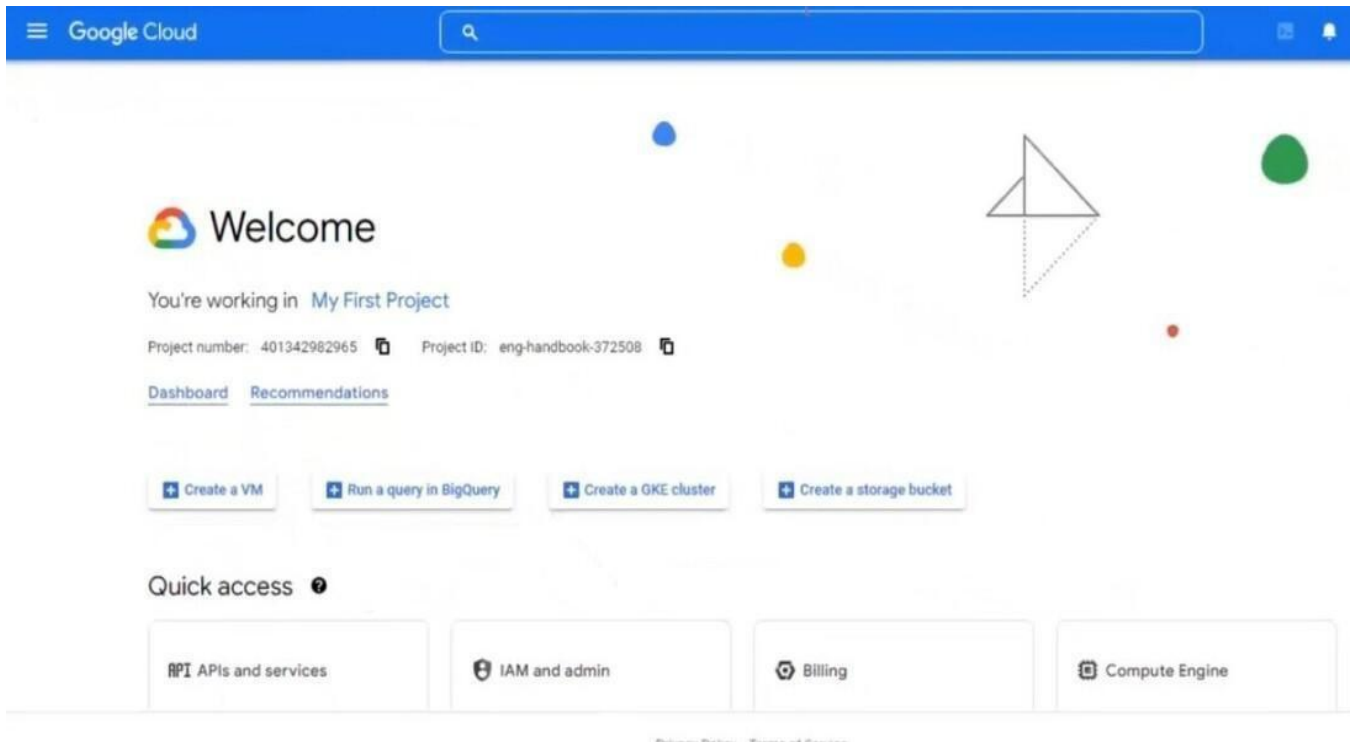
Google Kubernetes Engine (GKE) is a managed Kubernetes service provided by GCP. It simplifies cluster management, upgrades, monitoring, and security. Using GKE offers several benefits:

- Easy provisioning and scaling of clusters.
- Built-in integrations with GCP services (like Cloud Monitoring, Cloud Storage, etc.).
- Automated updates and security patches.
- High availability and multi-zone deployments.

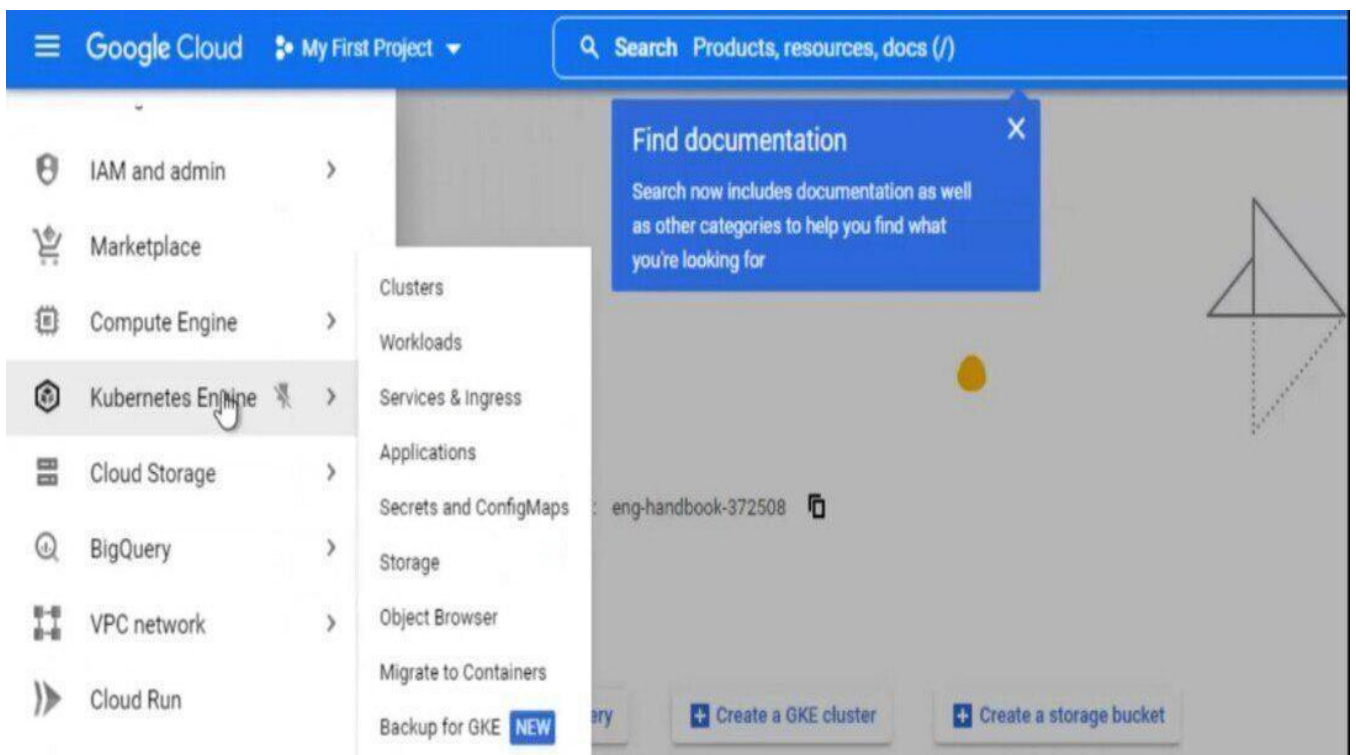


Implementation Workflow Overview

Step 1: Log in to Google Cloud Platform (<https://console.cloud.google.com/>)



Step 2: Go to the navigational menu and select Kubernetes Engine.





Step 3 : Enable Kubernetes API by clicking on Enable button.

Google Cloud My First Project

Product details

Kubernetes Engine API

Google Enterprise API

Builds and manages container-based applications, powered by the open source Kubernetes technology.

ENABLE TRY THIS API

Click to enable this API

OVERVIEW DOCUMENTATION

Step 4: Choose the Project in which you want to create Kubernetes Cluster.

Google Cloud My First Project Search: kuber

Kubernetes Engine

- Clusters
- Workloads
- Services & Ingress
- Applications
- Secrets and ConfigMaps
- Storage
- Object Browser
- Migrate to Containers

Notifications

- ✓ Create Project: gke-demo Just now
SELECT PROJECT
- ✓ Enable service: container.googleapis.com 1 minute ago
My First Project
- ✓ Create Project: My First Project 7 days ago
SELECT PROJECT

SEE ALL ACTIVITIES



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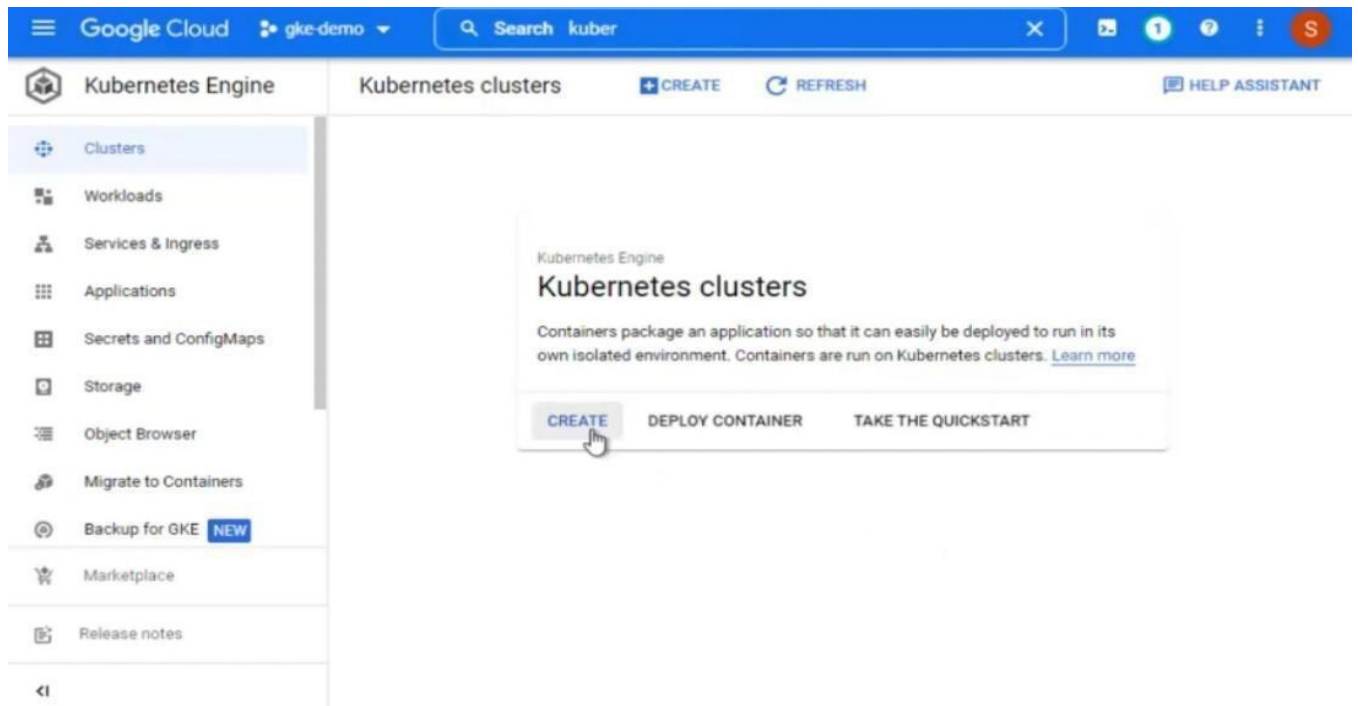
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Step 5: Now, after enabling API and choosing project its time to Create Kubernetes Cluster, Click on Create button.



Step 6: Give all the basic details like Cluster Name, Location, Zone etc and then click on the create button.



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Google Cloud

gke-demo

Search kuber

X

1

Create a Kubernetes cluster

Cluster basics

NODE POOLS

default-pool

CLUSTER

Automation

Networking

Security

Metadata

Features

Cluster basics

The new cluster will be created with the name, version and in the location that you specify here. After the cluster is created, name and location can't be changed.

To experiment with an affordable cluster, try **My first cluster** in the **Cluster set-up guides**

Name

gke-demo-cluster

Cluster names must start with a lowercase letter followed by up to 39 lowercase letters, numbers or hyphens. They can't end with a hyphen. You cannot change the cluster's name once it has been created.

Location type

Resource prices may vary between certain regions. [Learn more](#)

☒ Zonal

☐ Regional

Zone

us-central1-c

CREATE

CANCEL

Equivalent

REST

or

COMMAND LINE

Google Cloud

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X

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?

Create a Kubernetes cluster

Cluster basics

NODE POOLS

default-pool

Nodes

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Security

Metadata

CLUSTER

Automation

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Security

Current default: us-central1-c

Control plane version

Choose whether you'd like to upgrade the cluster's control plane version manually or let GKE do it automatically. [Learn more](#)

☐ Static version

Manually manage the version upgrades. GKE will only upgrade the control plane and nodes if it's necessary to maintain security and compatibility, as described in the release schedule. [Learn more](#)

☒ Release channel

Let GKE automatically manage the cluster's control plane version. [Learn more](#)

Release channel

Regular channel (default)

Version

1.24.7-gke.900 (default)

These versions have passed internal validation and are considered production quality but don't have enough historical data to guarantee their stability. Known issues generally have known workarounds. [Release notes](#)

CREATE

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Equivalent

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or

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Kubernetes is Successfully deploy in Google Cloud Platform (GCP).

Conclusion

In this study, we explored the theoretical foundations and practical implementation of container orchestration using Kubernetes on Google Cloud Platform. Kubernetes simplifies the deployment and management of containerized applications by providing robust orchestration features like auto-scaling, load balancing, and self-healing. Utilizing Google Kubernetes Engine further abstracts infrastructure management, enabling developers to focus on application logic rather than operational overhead.

By successfully deploying applications on GKE, we demonstrated how cloud-native technologies can improve scalability, reliability, and maintainability of modern applications. This project lays a strong foundation for further exploration into CI/CD pipelines and advanced Kubernetes features like Helm, RBAC, and persistent storage management.